

观点4更正（Corrigendum）

Cercis-Fisher等价性：条件等价而非自然涌现

Cercis-Fisher Equivalence: Conditional Equivalence, Not Natural Emergence

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2026 年 7 月 2 日

1 审查裁决 / Review Verdict

原主张 / Original Claim	裁决 / Verdict
$\text{Cercis}^2 \propto \text{平均KL散度} = \text{Fisher测地线长度平方}$ 。从信息几何自然涌现。	数学关系正确但“自然涌现”夸大 / Mathematically correct but “natural emergence” is overstated

1.1 具体问题 / Specific Issues

- (1) 不是“平均KL”而是“最小KL”。Cercis是到纯规范子流形的最小KL散度： $\mathcal{C} \approx 2 \cdot \min_h \text{KL}(P_A \| P_{d_{0h}})$ ，而非“平均KL散度”。

Not “average KL” but “minimum KL”. Cercis is the *minimum* KL divergence to the pure-gauge submanifold, not an average.

- (2) 不是“自然涌现”而是条件等价。GeoDist² ↔ 2KL 的等价性依赖8个强假设（见第2节）。任何一个假设不满足即断裂。SCX数据极不可能恰好形成指数族。

Not “natural emergence” but conditional equivalence. The GeoDist² ↔ 2KL equivalence depends on 8 strong assumptions (see §2). If any single assumption fails, the equivalence breaks. SCX data is extremely unlikely to form an exponential family.

- (3) 非指数族下完全失效。对一般统计流形，GeoDist²与2KL的差由Amari-Chentsov三阶张量控制。该张量在一般SCX数据上非零，等价性由此退化。

Complete breakdown outside exponential families. For general statistical manifolds, the difference between GeoDist² and 2KL is controlled by the Amari-Chentsov third-order tensor. This tensor is generically non-zero for SCX data, so the equivalence degrades.

2 八大假设清单 / The Eight Assumptions

要使得 $\text{GeoDist}(p, q)^2 = 2\text{KL}(p \| q) + O(\|p - q\|^3)$ 严格成立，以下八个假设必须同时满足。任何一个的失效都会导致等式退化。

For $\text{GeoDist}(p, q)^2 = 2\text{KL}(p||q) + O(\|p - q\|^3)$ to hold rigorously, the following eight assumptions must be simultaneously satisfied. Failure of any single one degrades the equality.

H1: 指数族假设 (Exponential Family Assumption).

中文: 输出分布族 $\mathcal{P} = \{p(x | \theta)\}$ 必须构成指数族, 即存在充分统计量 $T(x)$ 和累积量生成函数 $\psi(\theta)$ 使得 $p(x | \theta) = h(x) \exp(\langle \theta, T(x) \rangle - \psi(\theta))$ 。只有在此条件下, Fisher信息矩阵才是 ψ 的Hessian, KL散度才等于Bregman散度 (引理4.3 / Lemma ??的原上下文)。

English: The output distribution family $\mathcal{P}$ must be an exponential family. Only then is the Fisher matrix the Hessian of ψ , and KL divergence equals Bregman divergence.

断裂后果: 非指数族下, $\text{KL}(\theta_1||\theta_2)$ 不再是 ψ 的Bregman散度, Taylor展开的第二项不再是 $g_{ij}\Delta\theta^i\Delta\theta^j$ 。GeoDist² 与 2KL 的差由 Amari-Chentsov 三阶张量控制 (开放问题6.2 / Open Problem ??)。

Consequence of failure: Outside exponential families, KL is no longer a Bregman divergence, and the second-order Taylor term is no longer $g_{ij}\Delta\theta^i\Delta\theta^j$. The difference is controlled by the Amari-Chentsov third-order tensor (Open Problem ??).

H2: 小距离 / 局部近似假设 (Small-Distance / Local Approximation Assumption).

中文: θ_p 与 θ_q 必须充分接近, 使得三阶及以上余项 $O(\|\theta_q - \theta_p\|^3)$ 可忽略。该等式的所有推导均基于Taylor展开截断至二阶。

English: θ_p and θ_q must be sufficiently close so that third-order and higher remainders $O(\|\theta_q - \theta_p\|^3)$ are negligible. All derivations truncate the Taylor expansion at second order.

断裂后果: 在大分歧 (large discrepancy) 情形下, 三阶误差项主导, $\text{GeoDist}^2 \approx 2\text{KL}$ 不再成立。SCX跨领域专家的大分歧是常见现象。

Consequence of failure: For large discrepancies (common in cross-domain SCX experts), third-order error terms dominate and $\text{GeoDist}^2 \approx 2\text{KL}$ no longer holds.

H3: C^2 可微性假设 (C^2 Differentiability Assumption).

中文: 映射 $\theta \mapsto p(x | \theta)$ 必须对几乎所有 x 是 C^2 的。这是Taylor展开至二阶、以及Fisher信息矩阵作为二阶导数的定义的前提。

English: The map $\theta \mapsto p(x | \theta)$ must be C^2 for almost all x . This is required for second-order Taylor expansion and for the Fisher matrix as a second derivative.

断裂后果: 如果密度函数包含不可微点 (如ReLU激活产生的分段分布), Fisher度量在这些点不定义, 测地线无法唯一确定。

Consequence of failure: If the density has non-differentiable points (e.g., piecewise distributions from ReLU activations), the Fisher metric is undefined at those points and geodesics are not uniquely determined.

H4: 积分-微分可交换假设 (Integration-Differentiation Interchangeability Assumption).

中文: $\partial_\theta \int p(x|\theta)d\mu(x) = \int \partial_\theta p(x|\theta)d\mu(x)$ 必须成立。这是Fisher信息矩阵的标准正则性条件, 确

保 $\mathbb{E}[\partial_\theta \log p] = 0$ 。

English: The interchange $\partial_\theta \int p = \int \partial_\theta p$ must hold. This is the standard regularity condition ensuring $\mathbb{E}[\partial_\theta \log p] = 0$, which underpins the Fisher information definition.

断裂后果: 若支撑集依赖于参数 (如均匀分布), Fisher信息的标准形式不再成立。

Consequence of failure: If the support depends on θ (e.g., uniform distributions), the standard Fisher information form breaks down.

H5: Fisher满秩假设 (Fisher Full-Rank Assumption).

中文: Fisher信息矩阵 $\mathcal{I}(\theta)$ 必须在每一点正定 (满秩)。这是 Θ 上Fisher度量成为真正黎曼度量的必要条件——退化度量不定义唯一的测地线。

English: The Fisher information matrix $\mathcal{I}(\theta)$ must be positive-definite (full rank) at every point. This is required for (Θ, g) to be a genuine Riemannian manifold — a degenerate metric does not define unique geodesics.

断裂后果: 若参数间存在函数依赖 (如归一化约束导致的有效降维), Fisher矩阵奇异, 测地线不存在或不唯一, GeoDist 无定义。

Consequence of failure: If parameters are functionally dependent (e.g., normalization constraints causing effective dimensionality reduction), the Fisher matrix is singular, geodesics may not exist or be unique, and GeoDist is undefined.

H6: 小联络近似假设 (Small-Connection Approximation Assumption).

中文: 边赋值 a_e 与规范变换 h_v 必须在小联络极限内 ($\max_e \text{dist}_{O(d)}(A_e, I_d) < r_0$, 其中 r_0 为矩阵对数的收敛半径)。只有在此极限下, 非阿贝尔规范固定才退化为李代数 $\mathfrak{so}(d)$ 上的线性霍奇问题。

English: Edge assignments a_e and gauge transformations h_v must be within the small-connection limit ($\max_e \text{dist}_{O(d)}(A_e, I_d) < r_0$, the matrix logarithm convergence radius). Only then does non-abelian gauge-fixing reduce to linear Hodge theory on $\mathfrak{so}(d)$.

断裂后果: 大旋转下 (如跨模态专家间的90°标架差异), Cartan分解非线性项不能忽略, GeoDist与欧氏距离的关系断裂。

Consequence of failure: Under large rotations (e.g., 90° frame misalignment between cross-modal experts), nonlinear Cartan terms cannot be neglected, and the relationship between GeoDist and Euclidean distance breaks.

H7: 欧氏近似假设 (Euclidean Approximation / Degenerate Fisher Assumption).

中文: Cercis的欧氏定义 ($\|r\|^2 = \|r_{\text{harm}}\|^2 + \|r_{\text{coexact}}\|^2$) 要与Fisher定义等价, 需要Fisher度量退化为欧氏度量: $\mathcal{I}_{ij}(\theta) = \delta_{ij}$ 。即信息流形的所有方向权重相等, 曲率为零。

English: For the Euclidean Cercis definition ($\|r\|^2 = \|r_{\text{harm}}\|^2 + \|r_{\text{coexact}}\|^2$) to coincide with the Fisher definition, the Fisher metric must degenerate to Euclidean: $\mathcal{I}_{ij}(\theta) = \delta_{ij}$. This means all directions on the information manifold have equal weight and zero curvature.

断裂后果： 真实SCX数据的参数空间几乎肯定具有非平凡的Fisher曲率。Cercis的欧氏范数和Fisher测地距离之间存在不可忽略的曲率修正。

Consequence of failure: Real SCX parameter spaces almost certainly have non-trivial Fisher curvature. A non-negligible curvature correction exists between Euclidean Cercis and Fisher geodesic distance.

H8: 分布紧致假设 (Distribution Compactness / Bounded Support Assumption).

中文： 所有专家输出分布必须在 Θ 的紧致子集上充分集中，使得局部Taylor展开在边界观测和所有规范体态上一致有效。

English: All expert output distributions must be sufficiently concentrated on a compact subset of Θ , so that the local Taylor expansion is uniformly valid across all boundary observations and gauge bulk states.

断裂后果： 重尾分布（如学生-t分布）可能使某些观测远离局部有效域，导致三阶误差项在个别边上不可控地增长。

Consequence of failure: Heavy-tailed distributions (e.g., Student's-t) may place some observations outside the local validity domain, causing uncontrolled growth of third-order error terms on individual edges.

3 修正后的主张 / Corrected Claim

3.1 中文修正 (Chinese Correction)

原主张（夸大，需删除）：

“Cercis分数是Fisher信息度量下的测地线距离。指数族假设下 $\text{Cercis}^2 \propto \text{平均KL散度} = \text{Fisher测地线长度平方}$ 。审计分数不是人为定义——从数据流形信息几何自然涌现。”

修正后的主张（条件等价）：

在指数族假设和局部近似下，Cercis分数等于观测边分布到纯规范子流形的最小KL散度的两倍，而后者又等于（至二阶）Fisher测地线距离的平方。这提供了一种信息几何解释——但并非“自然涌现”，而是条件等价：当且仅当Situs流形承认指数族结构时成立。对于非指数族数据，该关系由Amari-Chentsov三阶张量退化（开放问题6.2）。

具体表述为：

定理（Cercis的信息几何条件等价性，修订版）

假设：上述8个假设(H1–H8)同时成立。

结论：

$$\mathcal{C} = \min_{d_0 h \in \text{im}(d_0)} \text{GeoDist}(P_A, P_{d_0 h})^2 = 2 \min_{d_0 h \in \text{im}(d_0)} \text{KL}(P_A \| P_{d_0 h}) + O(\|\mathbf{a}\|^3).$$

Cercis度量了从观测专家边缘分布到“纯规范体态”（可完美对齐的假想世界）的最小Fisher信息距离。

限制：

- (i) 该等价性在指数族(H1)内严格成立；非指数族下， GeoDist^2 与 2KL 的差由Amari-Chentsov三阶张量 $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$ 控制，且 T_{ijk} 在一般SCX数据上非零（开放问题??）。
- (ii) 该等价性在局部(H2)近似下成立；大分歧下三阶项主导。
- (iii) “自然涌现”是错误的描述——这是一组强假设下的构造性等价，而非数据固有的几何必然性。

3.2 English Correction

Original claim (overstated, to be removed):

“The Cercis Score is the geodesic distance under the Fisher information metric. Under the exponential family assumption, $\text{Cercis}^2 \propto \text{mean KL divergence} = \text{squared Fisher geodesic length}$. The audit score is not artificially defined — it naturally emerges from the information geometry of the data manifold.”

Corrected claim (conditional equivalence):

Under the exponential family assumption and local approximation, the Cercis Score equals twice the **minimum** KL divergence from the observed edge distribution to the pure-gauge submanifold, which in turn equals (to second order) the squared Fisher geodesic distance. This provides an information-geometric **interpretation** — not a “natural emergence” but a **conditional equivalence** that holds when and only when the Situs manifold admits an exponential family structure. **For non-exponential-family data, the relationship degrades by the Amari-Chentsov tensor (Open Problem 6.2).**

Explicit formulation:

Theorem (Information-Geometric Conditional Equivalence of Cercis, Revised)**Assumptions:** All 8 assumptions (H1–H8) hold simultaneously.**Conclusion:**

$$\mathcal{C} = \min_{d_0 h \in \text{im}(d_0)} \text{GeoDist}(P_A, P_{d_0 h})^2 = 2 \min_{d_0 h \in \text{im}(d_0)} \text{KL}(P_A \| P_{d_0 h}) + O(\|\mathbf{a}\|^3).$$

Cercis measures the minimal Fisher information distance from the observed expert edge distributions to the “pure-gauge bulk states” — the hypothetical world where all experts can be perfectly aligned.

Limitations:

- (i) The equivalence is rigorous within exponential families (H1); outside them, the difference between GeoDist^2 and 2KL is controlled by the Amari-Chentsov third-order tensor $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$, which is generically non-zero for SCX data (Open Problem ??).
- (ii) The equivalence holds under local approximation (H2); for large discrepancies, third-order terms dominate.
- (iii) “Natural emergence” is an incorrect characterization — this is a *constructed equivalence* under strong assumptions, not an intrinsic geometric inevitability of the data.

4 五大断裂点 / Five Breakdown Points

以下总结了等价性在实际SCX数据上的五个主要断裂途径：

Below, we summarize five principal ways the equivalence can break on real SCX data:

B1: 非指数族失效（最严重） / Non-Exponential-Family Failure (Most Severe).

SCX专家输出（特别是MoE路由权重、softmax得分、注意力分布）极不可能恰好构成指数族。指数族要求特定的代数结构（线性充分统计量 $T(x)$ ），而深度学习输出通常不满足。文档自身将其列为开放问题6.2（??）。

SCX expert outputs (MoE routing weights, softmax scores, attention distributions) are extremely unlikely to form an exponential family. Exponential families require specific algebraic structure (linear sufficient statistics $T(x)$) that deep learning outputs do not generally satisfy. The document itself lists this as Open Problem 6.2 (??).

B2: 混淆“最小KL”与“平均KL” / Confusing “min KL” with “average KL”.

Cercis是最小化问题 $\min_h \text{KL}(P_A \| P_{d_0 h})$ 的解，不是对所有专家对取平均的KL散度。这两个概念在数学上有本质区别。

Cercis solves $\min_h \text{KL}(P_A \| P_{d_0 h})$ — a minimization problem, not an average of KL divergences over all expert pairs. These are mathematically distinct.

B3: 欧氏近似的隐性要求 / Hidden Euclidean Approximation Requirement.

Cercis的欧氏范数定义与Fisher测地距离的等价性要求 $\mathcal{I}_{ij} = \delta_{ij}$ (H7)。这意味着信息流形必须处处平坦——在实际的SCX参数空间中极不可能成立。

The equivalence between Euclidean-norm Cercis and Fisher geodesic distance requires $\mathcal{I}_{ij} = \delta_{ij}$ (H7) — meaning the information manifold must be everywhere flat, which is extremely unlikely on real SCX parameter spaces.

B4: 定理4.2证明中的内在修正 / Intrinsic Correction in Theorem 4.2 Proof.

选错了子流形会丢失调和分量。正确定义为：体态子流形由正合联络（纯规范） $\text{im}(d_0)$ 参数化，而非调和体态 $\ker(L_1)$ 。原定理??在证明中发现并修正了此问题。

Choosing the wrong submanifold loses the harmonic component. The correct definition is: the bulk submanifold is parameterized by exact connections (pure gauge) $\text{im}(d_0)$, not harmonic bulk states $\ker(L_1)$. Theorem ?? identifies and corrects this issue in its proof.

B5: 局部性限制 / Locality Constraint.

所有结果仅在 $\|\theta_q - \theta_p\|$ 充分小时成立。当跨构型的专家差异较大（如不同领域、不同训练阶段的专家），三阶误差项 $O(\|\Delta\theta\|^3)$ 不可忽略。

All results hold only when $\|\theta_q - \theta_p\|$ is sufficiently small. When inter-configuration expert differences are large (e.g., different domains, different training stages), the third-order error term $O(\|\Delta\theta\|^3)$ is non-negligible.

5 修订建议 / Recommended Revisions

5.1 在主文件中需要的修改 / Changes Needed in gauge_formalized.tex

- (1) 摘要（第234–238行）：删除“自然涌现”相关措辞。将

“我们证明在此族下，Fisher测地线长度与KL散度满足局部等价性...”

改为：

“我们证明在指数族假设和局部近似下，Fisher测地线长度与KL散度之间存在条件等价关系...”

英文摘要同理修改。

- (2) 第??节标题：将

“指数族下的Fisher测地线与KL散度的等价性 / Fisher Geodesic–KL Equivalence under Exponential Families”

改为：

“指数族下的Fisher测地线与KL散度的条件等价性 / Conditional Fisher Geodesic–KL Equivalence under Exponential Families”

(3) 定理??：在定理陈述之前添加明确的假设列表，引用上述H1–H8。在定理结尾添加限制性注记。

(4) 定理??（统一结构定理）：将措辞从

“Cercis重新表述为从边界到纯规范体态的Fisher测地距离，在指数族下局部等价于2KL”

改为：

“在指数族假设和局部近似下，Cercis可重新解释为从边界到纯规范体态的Fisher测地距离，条件等价于2KL（至二阶）。非指数族下的行为由开放问题6.2处理。”

(5) 比较表（第1529–1532行）：将

“Cercis定义: $\min_{d_{0h}} \text{GeoDist}(P_A, P_{d_{0h}})^2$ ” 和 “Cercis概率解释: $\approx 2 \min_{d_{0h}} \text{KL}(P_A \| P_{d_{0h}})$ ”

添加脚注或括号注：“（需指数族假设，非指数族下退化）”

(6) 第??节“信息几何的严格性”段落（第1560–1564行）：将

“本文将其应用于SCX的S流形是自然的推广...”

改为：

“本文将其应用于SCX的S流形是在特定假设下的条件推广。当这些假设不满足时（特别非指数族情形，开放问题6.2），等价性由Amari-Chentsov张量退化。”

(7) 结论第3点（第1678–1684行）：将

“证明了指数族下Fisher测地线长度与KL散度的局部等价性...建立了Cercis分数的信息几何解释”

改为：

“证明了指数族假设下Fisher测地线长度与KL散度的条件等价性...建立了Cercis分数的条件信息几何解释，并明确标注了该解释的8个前提假设和失效条件。”

6 开放问题6.2的交叉引用 / Cross-Reference to Open Problem 6.2

本更正与开放问题??（“非指数族下的Fisher-KL等价性”）直接关联。该问题指出：

对于一般的SCX输出分布（可能非指数族），Fisher测地距离和KL散度之间的关系由 Amari-Chentsov 三阶张量 $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$ 控制。需要该张量在SCX输出的经验分布上的界。

This corrigendum is directly linked to Open Problem ?? (“Fisher-KL Equivalence Outside Exponential Families”), which states:

For general SCX output distributions (potentially non-exponential-family), the relationship between Fisher geodesic distance and KL divergence is controlled by the Amari-Chentsov third-order tensor $T_{ijk} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p \cdot \partial_k \log p]$. Bounds on this tensor for empirical SCX output distributions are needed.

审查依据：本更正基于以下审查文件：`docs/reviews/GAUGE_5VIEWPOINTS_FINAL.md`（裁决：“夸大”）及 `docs/reviews/GAUGE_VIEWPOINTS_REVIEW.md`（详细分析）。对观点4的裁决为✓数学关系正确但“自然涌现”严重夸大，需降级为“条件等价”。

Review basis: This corrigendum is based on: `docs/reviews/GAUGE_5VIEWPOINTS_FINAL.md` (verdict: “overstated”) and `docs/reviews/GAUGE_VIEWPOINTS_REVIEW.md` (detailed analysis). The verdict on Viewpoint 4 is: ✓ mathematically correct relationship but “natural emergence” is severely overstated — downgrade to “conditional equivalence”.